Quick Guide: Modbus 4-20mA Distance and Level Sensor

1. Main Technical Parameters

- **Measurement range:** 0.2…8 m (hardware "blind zone" on receiver is less than 200 mm).
- **Measurement resolution:** 1 mm (due to interpolation in the built-in filter).
- **Supply voltage:** 8…28 V direct current (VDC).
- **Power consumption:** not more than 3 W.
- **Communication interface:** RS-485 (two-wire, half-duplex, default port parameters: 9600 bps, 8N1).
- **Analog output:** active 4–20 mA current loop (internally powered, 12-bit DAC, load resistance not more than 250 Ohm).
- **Analog output modes:** 0 — by distance (mm), 1 — by fill level (%), 2 — discrete alarm (dry contact).

2. Electrical Connection (Connector Pinout and Wire Colors)

The device must be connected with the power disconnected. For long communication lines, it is recommended to install a 120 Ohm terminating resistor at the ends of the RS-485 line.

Pin	Wire Color	Signal	Purpose
1	Red	8...28VDC	Power supply plus (+)
2	Blue	0V	Power supply minus / Common (-)
3	Black	RS485-A	RS-485 communication line (A / RxD+)
4	Yellow	RS485-B	RS-485 communication line (B / TxD-)
5	Green	AO+	Current loop output (+)
6	White	AO-	Current loop output (-)

3. Modbus Register Map (Holding Registers)

Data exchange is performed using standard Modbus RTU functions: 03h (read), 06h, and 10h (write). Parameter changes are applied immediately after writing.

Registers	Parameter	Access	Data Type	Description / Default Values
0, 1	Measurements	R	UInt16	**Reg0:** Distance (mm). **Reg1:** Reflected signal level from the measurement module (arb. units).
2, 3	Measurements	R	Int16 / UInt16	**Reg2:** Sensor temperature (°C, signed). **Reg3:** Calculated output current (hundredths of mA, 400 = 4.00 mA).
4, 5	Bin Status	R	UInt16	**Reg4:** Fill level (%) [0..100]. **Reg5:** Discrete status (0 — normal, 1 — filled).

Registers	Parameter	Access	Data Type	Description / Default Values
6	Errors	R	UInt16	**Reg6:** System error status (0 — normal, Bit 0 — rangefinder failure, Bit 1 — DAC failure).
10, 11	Comm / Polling	R/W	UInt16	**Reg10:** Modbus address (1..247, default: 1). **Reg11:** Measurement frequency (1..100 Hz, default: 10).
12, 13	Filtering	R/W	UInt16	**Reg12:** Median window (1..15 samples, default: 10). **Reg13:** EMA smoothing (0..100%, default: 20%).
14, 15	Calibration	R/W	UInt16	Coordinates of reference points **D1** (Reg14, default: 500 mm) and **D2** (Reg15, default: 5000 mm).
16, 17	Current Range	R/W	UInt16	Distance boundaries **X1** for 4 mA (Reg16, default: 250) and **X2** for 20 mA (Reg17, default: 5000).
18, 19	Output DAC	R/W	UInt16	Calibration DAC codes **Y1** for 4 mA (Reg18, default: 819) and **Y2** for 20 mA (Reg19, default: 4095).
20	Operating Mode	R/W	UInt16	**0:** operation. **1, 2:** DAC calibration. **3, 4:** sensor calibration. **5, 6:** empty / full bunker capture.
21, 22	Bunker Boundaries	R/W	UInt16	**Reg21:** Empty bin bottom D_empty (default: 5000). **Reg22:** Full bin top D_full (default: 500).
23	DAC Mode	R/W	UInt16	Analog output mode: 0 — Distance, 1 — Level (%), 2 — Alarm. (default: 0).
24, 25	Level Control	R/W	UInt16	**Reg24:** Alarm threshold (10..100%, default: 90). **Reg25:** Hysteresis (1..50%, default: 5).

4. Calibration of the Sensor and Bunker Boundaries

Built-in protection algorithms automatically maintain the mathematical integrity of calibrations, preventing calculator freeze on accidental operator error (minimum range delta is 10 mm).

1. **Sensor Calibration (D1/D2):**
   - Place the reflector in front of the sensor at a stable, known distance (e.g., 500 mm), write this value to `Reg14 = 500`, and start the capture: `Reg20 = 3`.
   - Move the reflector to the second known distance (e.g., 5000 mm), write it to `Reg15 = 5000`, and start the capture: `Reg20 = 4`. The device will calculate the coefficients and save them.
2. **Automatic Capture of Container Boundaries (D_empty/D_full):**
   - *Empty container (0%):* Empty the container and write the command `Reg20 = 5`. The current filtered distance will be automatically saved in `Reg21` as the bottom of the bin.
   - *Full container (100%):* Fill the container to maximum and write the command `Reg20 = 6`. The current distance to the material will be saved in `Reg22` as the upper technological limit.
3. Write the command `Reg20 = 0` to the control register to return the device to normal operating mode.

5. Configuration and Calibration of the 4–20 mA Analog Output

1. **Mode Selection (Reg23):** Set the current loop mode. **0** — proportional to distance (X1..X2 in mm); **1** — proportional to fill level (0..100% from D_empty to D_full); **2** — discrete alarm (4 mA for empty, 20 mA for overfill based on the thresholds in `Reg24` and `Reg25`).
2. **Scaling (for Mode 0):** Specify the distance range in registers `Reg16` (X1 — distance for 4 mA) and `Reg17` (X2 — distance for 20 mA).
3. **Current Loop Calibration (Y1/Y2 Correction):** Connect a milliammeter in series to the analog output. Ensure that the load resistance in the current loop does not exceed 250 Ohm.
   - *4 mA Calibration:* Write the command `Reg20 = 1`. Adjust the value in register `Reg18` until the milliammeter shows exactly 4.00 mA.
   - *20 mA Calibration:* Write the command `Reg20 = 2`. Adjust the value in register `Reg19` until the milliammeter displays exactly 20.00 mA.
4. **Completion:** Write the command `Reg20 = 0` to return the device to normal operating mode.

6. Installation and Operation Requirements

- To achieve maximum measurement accuracy, the laser beam must be directed strictly perpendicularly to the surface of the object (for liquid level monitoring, the device should be mounted strictly over the center of the reservoir).
- Avoid the beam intersecting with foreign objects in the working zone (internal ladders, pipes, partitions). Note that the laser cone diverges at an angle of **2° FoV**. The light spot diameter increases by approximately 3.5 cm for every 1 meter of distance (at a distance of 8 m, the diameter will be 28 cm).
- Reflectivity directly affects the maximum range: for light-colored surfaces, the range is up to **8.0 m**; for black absorbing materials (e.g., coal or dark wet backfills), the range drops to **2.5 m**.
- Prevent condensation and heavy contamination from forming on the optical window.

7. Troubleshooting Table

Fault	Possible Cause	Remedy
No Modbus communication	No power / Swapped communication lines A and B / Incorrect address or port parameters	Check power supply (8...28 V); swap communication wires A and B; verify address in `Reg10`; configure master device to speed 9600, frame format 8N1.
Register 0 is always 0	Operation in "blind zone" (< 200 mm) / Dirty optics	Increase distance to object; clean the protective window with a soft cloth without abrasives.
Fill level (Reg4) is always 0%	D_empty (Reg21) is less than or equal to D_full (Reg22)	Check and adjust settings in `Reg21` and `Reg22`, then restart empty/full container capture (Section 4, step 2).
Output current is frozen at 4 mA	Distance is less than X1 / Device remains in calibration mode	Verify values in `Reg16` and `Reg17`; ensure that 0 is written to the control register `Reg20`.

Fault	Possible Cause	Remedy
Output current does not match the distance or is constantly 4 mA when connection exists	SCL/SDA bus error or DAC hardware hang-up	Check the value of the error register `Reg6`. If Bit 1 (value 2) is set — check the integrity of PCB traces to the DAC microchip, the reliability of its soldering, and the absence of short circuits on SDA/SCL lines. Re-calibrate the current output.